

SHREYA SUTAR

(CALIFORNIA ACADEMY OF MATHEMATICS AND SCIENCE, JUNIOR)

SKILLS

Mechanical: CAD (Fusion 360, Inventor, Onshape — 200+ hours); technical, fabrication, part, and assembly drawings to ASME Y14.5-2012; GD&T, decision matrices, mechanism design, tendon-driven actuation, FDM 3D printing and design for manufacturability.

Electrical & Embedded: Circuit design and soldering, RP2040/Raspberry Pi Pico, MicroPython firmware, closed-loop control, power management, sensor integration (surface EMG, INA219 current sensing, IMU, servo drivers).

Computational & Research: Python, biosignal processing, gesture classification, data analysis, technical writing, peer review.

EXPERIENCE

EMG-Driven Assistive Exoskeleton Glove — Independent Research (ISEF 2027 Project)

SEPTEMBER 2026 - PRESENT

- Designing a tendon-driven exoskeleton glove for people with weak grip, built around a powered 2-DOF thumb, which most assistive gloves leave passive, to handle flat, delicate, and opposition-requiring objects.
- Developing a four-motor architecture: forearm-mounted gearmotors with encoders driving Bowden cables, finger pairs coupled through a floating-pulley differential, and magnetic encoders on side hinges aligned to the knuckle axis.
- Setting up multi-channel surface EMG acquisition for intent detection, with motor-current sensing for force-aware grasping.

Myoelectric Prosthetic Hand (PandaWare) — Independent Research

OCTOBER 2025 - PRESENT

- Independently designed, built, and self-funded a surface EMG-controlled prosthetic hand: mechanics, circuitry, and firmware.
- Developed a custom adaptive EMG normalization pipeline on an RP2040 in MicroPython, covering raw signal acquisition, feature extraction, and closed-loop servo control through a PCA9685 driver.
- Used an INA219 current sensor as a low-cost force proxy so the hand detects contact and stops before crushing an object.
- Engineered the mechanical design in CAD over multiple printed iterations: tendon routing, joint kinematics, moment arms, friction.
- Designing a second-generation hand with independent per-joint actuation (MCP, PIP, DIP) and passive spring return, to reach hand configurations that coupled tendon designs cannot produce.
- Wrote a first-author manuscript on the design and evaluation of the hand.

Founder & Editor-in-Chief, ISE — Innovation in Science and Engineering Journal

SEPTEMBER 2025 - PRESENT

- Founded and run an ISSN-registered student engineering journal publishing high school and undergraduate work.
- Built the journal's full manuscript infrastructure, including four IEEE-format templates (research, engineering, literature review, and computational) with formal heading hierarchies and back matter following Nature and PLOS conventions.
- Recruited advisors and reviewers through outreach to 30+ researchers nationwide, securing Dr. Ellen Roche (MIT) as faculty advisor, who now refers student submissions from national summer research programs.
- Manage submissions, review coordination, outreach, and the journal website; lead student engineering teams on hands-on projects.

Prosthetics Outreach Lead — Senior Living Community

JULY 2026 - PRESENT

- Lead a project bringing assistive technology to a senior living community, directing the CAD, electrical, and programming work.
- Write step-by-step lesson plans and build instructions so other students can contribute to the build rather than observe it.
- Demonstrate the device and help residents use it directly, translating technical work for non-technical users.

CAD Designer — FRC Team 687

JUNE 2025 - PRESENT

- Led CAD, prototyping, and testing of the Intake V5, Conveyor, and Indexer, delivering a faster, more reliable intake in one month.
- Integrated assembly, manufacturing, and programming feedback across subteams; team qualified for playoffs in the Newton Division at the FRC World Championship.

ADDITIONAL INFORMATION

Relevant Courses: Physics C: Mechanics, Calculus II, Aerospace Engineering. **SAT:** 1560.

Clubs & Activities: FIRST Robotics Competition (Team 687), ISE, SRC; Odissi classical dance (10+ years).

Awards: ISEF 2026 Nominee; CSEF 5th Place (Electronics and Electromagnetics); LACSEF 2nd Place (Electronics and Robotics); FRC Worlds Newton Division Playoffs; Invention Convention Grade Level Winner and Best Logbook Award.

Onshape CAD: <https://cad.onshape.com/documents/01846624b9cea47c0def8825/w/19642d9b3fb19bd6c92f9745/e/b12947016f13ac444aeb0029>